

1 (a) The total cost of a taxi journey is calculated as

- \$0.50 per kilometre
- plus
- \$0.40 per minute.

(i) Calculate the total cost of a journey of 32 km that takes 30 minutes.

\$ [2]

(ii) The total cost of a journey of 100 km is \$98.

Show that the time taken is 2 hours.

[3]

(b) Three taxi drivers travel a total of 8190 km in the ratio 5 : 2 : 7.

Calculate the distance each driver travels.

Driver 1 km

Driver 2 km

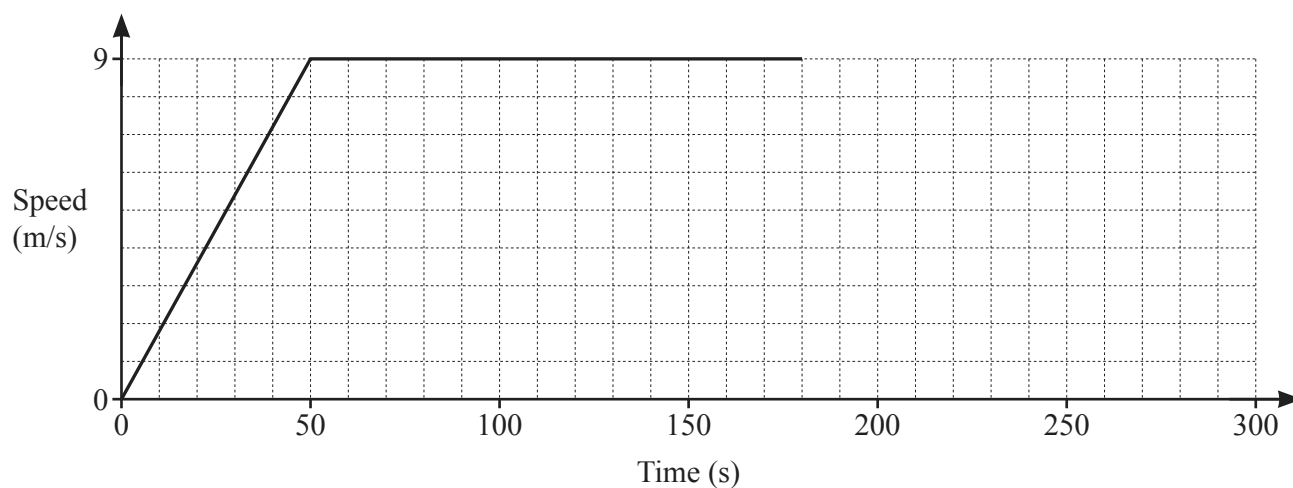
Driver 3 km [3]

(c) After midnight, the cost of any taxi journey increases by 45%.
One journey costs \$84.10 after midnight.

Calculate the cost of the same journey before midnight.

\$ [2]

- 2 The diagram shows the speed–time graph for the first 180 seconds of a train journey.



- (a) Find the acceleration, in m/s^2 , of the train during the first 50 seconds.

..... m/s^2 [1]

- (b) After 180 seconds, the train decelerates at a constant rate of 1944 km/h^2 .

Show that the train decelerates for 60 seconds until it stops.

[2]

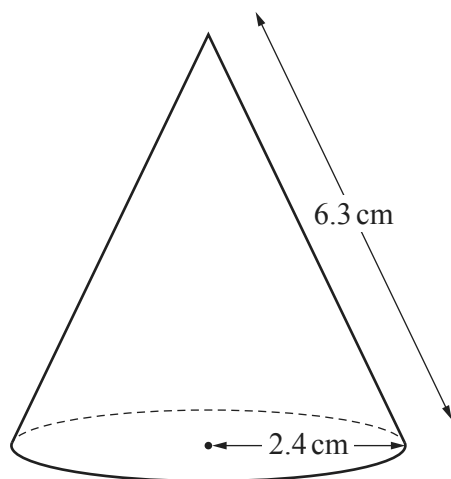
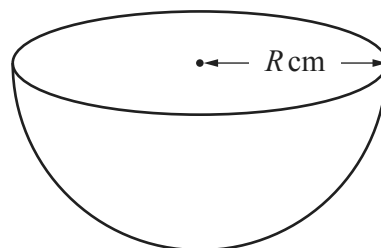
- (c) Complete the speed–time graph.

[1]

- (d) Calculate the average speed of the train for the whole journey.

..... m/s [4]

3 (a)

NOT TO
SCALE

The diagram shows a solid cone and a solid hemisphere.

The cone has radius 2.4 cm and slant height 6.3 cm.

The hemisphere has radius R cm.

The **total** surface area of the cone is equal to the **total** surface area of the hemisphere.

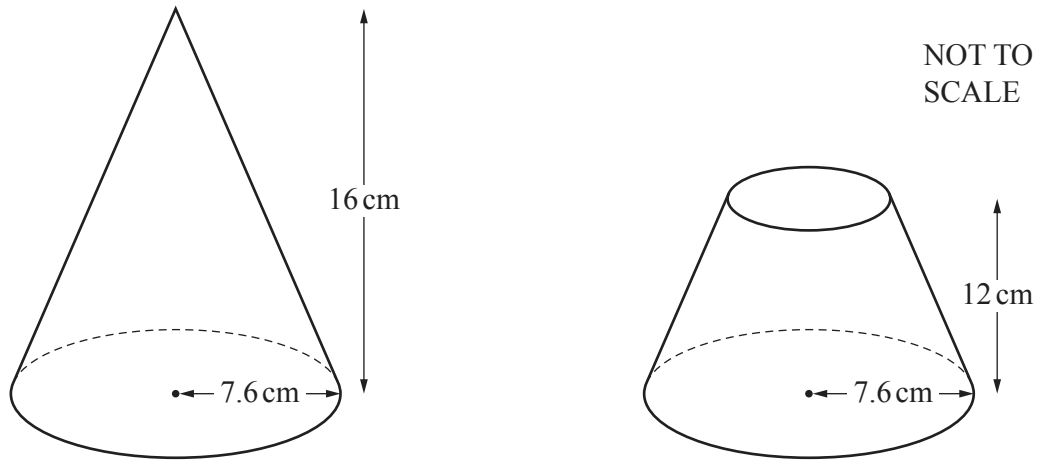
Calculate the value of R .

[The curved surface area, A , of a cone with radius r and slant height l is $A = \pi rl$.]

[The curved surface area, A , of a sphere with radius r is $A = 4\pi r^2$.]

$R =$ [4]

(b)



The diagram shows a solid cone with radius 7.6 cm and height 16 cm.
 A cut is made parallel to the base of the cone and the top section is removed.
 The remaining solid has height 12 cm, as shown in the diagram.

Calculate the volume of the remaining solid.

[The volume, V , of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

..... cm³ [4]

4 (a) The exchange rate is 1 euro = \$1.142 .

(i) Johann changes \$500 into euros.

Calculate the number of euros Johann receives.
Give your answer correct to the nearest euro.

..... euros [2]

(ii) Johann buys a computer for \$329.
The same computer costs 275 euros.

Calculate the difference in cost in dollars.

\$ [2]

(b) Lucy spends $\frac{3}{8}$ of the money she has saved this month on a book that costs \$5.25 .

Calculate how much money Lucy has saved this month.

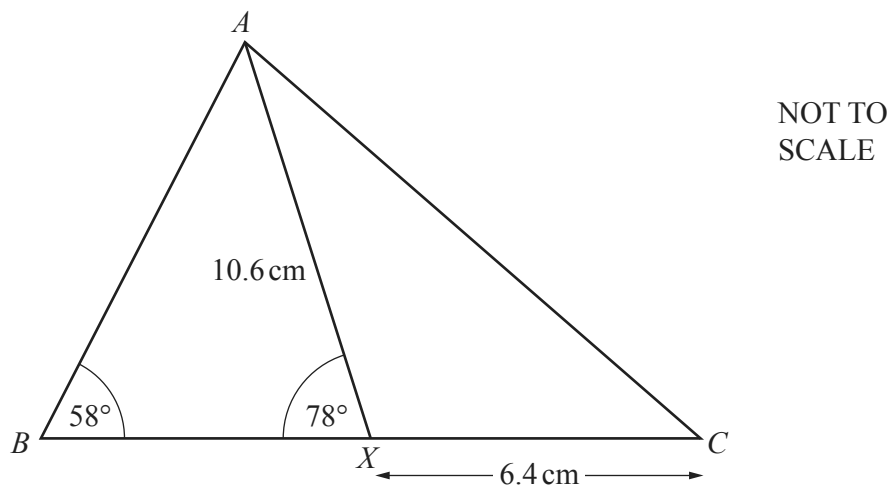
\$ [2]

(c) Kamal invests \$6130 at a rate of $r\%$ per year compound interest.
The value of his investment at the end of 5 years is \$6669.

Calculate the value of r .

$r =$ [3]

5



The diagram shows triangle ABC .

X is a point on BC .

$AX = 10.6\text{ cm}$, $XC = 6.4\text{ cm}$, angle $ABC = 58^\circ$ and angle $AXB = 78^\circ$.

(a) Calculate AC .

$AC = \dots\dots\dots\text{ cm}$ [4]

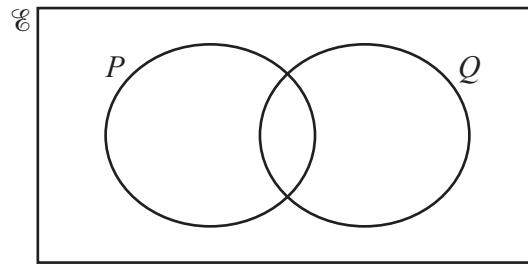
(b) Calculate BX .

$BX = \dots\dots\dots\text{ cm}$ [4]

(c) Calculate the area of triangle ABC .

$\dots\dots\dots\text{ cm}^2$ [3]

- 6 (a) In the Venn diagram, shade the region $P' \cup Q$.



[1]

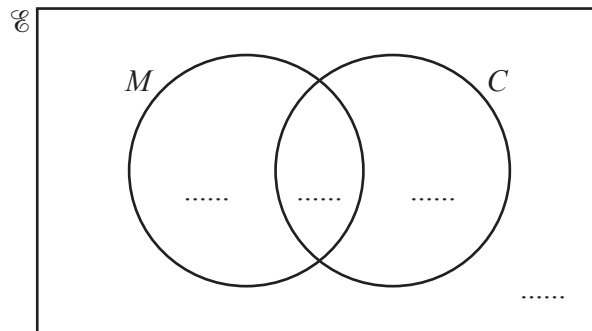
- (b) There are 50 students in a group.

34 have a mobile phone (M).

39 have a computer (C).

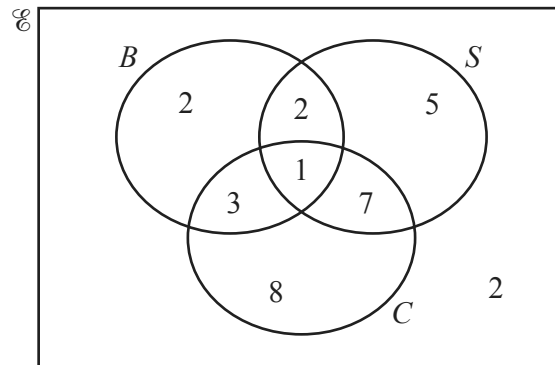
5 have no mobile phone and no computer.

Complete the Venn diagram to show this information.



[2]

- (c) The Venn diagram shows the number of students in a group of 30 who have brothers (B), sisters (S) or cousins (C).



- (i) Write down the number of students who have brothers.
 [1]
- (ii) Write down the number of students who have cousins but do not have sisters.
 [1]
- (iii) Find $n(B \cup S \cup C)$.
 [1]
- (iv) Use set notation to describe the set of students who have both cousins and sisters but do not have brothers.
 [1]
- (v) One student is picked at random from the 30 students.
 Find the probability that this student has cousins.
 [1]
- (vi) Two students are picked at random from the students who have cousins.
 Calculate the probability that both these students have brothers.
 [3]
- (vii) One student is picked at random from the 30 students.
 Event A This student has sisters.
 Event B This student has cousins but does not have brothers.
 Explain why event A and event B are equally likely.

 [1]

7 (a) Simplify.

$$\frac{x^2 - 25}{x^2 - x - 20}$$

..... [3]

(b) Write as a single fraction in its simplest form.

$$\frac{x+5}{x} + \frac{x+8}{x-1}$$

..... [3]

(c) A curve has equation $y = 2x^3 - 4x^2 + 6$.

(i) Find $\frac{dy}{dx}$, the derived function of y .

..... [2]

(ii) Calculate the gradient of the curve $y = 2x^3 - 4x^2 + 6$ at $x = 4$.

..... [2]

(iii) Find the coordinates of the two stationary points on the curve.

(.....,) and (.....,) [4]

- 8 (a) The table shows information about the mass, in kilograms, of each of 50 children.

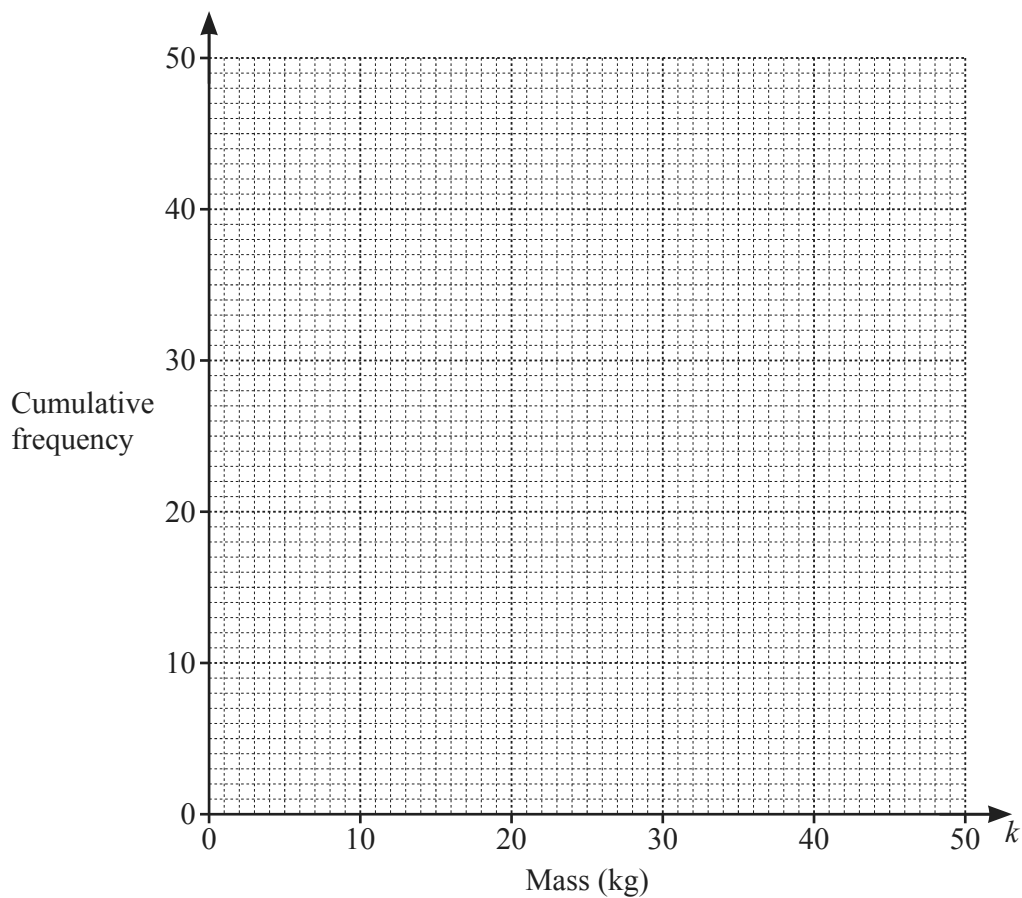
Mass (k kg)	$0 < k \leq 10$	$10 < k \leq 25$	$25 < k \leq 35$	$35 < k \leq 40$	$40 < k \leq 50$
Frequency	3	19	21	5	2

- (i) Complete the cumulative frequency table.

Mass (k kg)	$k \leq 10$	$k \leq 25$	$k \leq 35$	$k \leq 40$	$k \leq 50$
Cumulative frequency					

[2]

- (ii) On the grid, draw a cumulative frequency diagram to show this information.

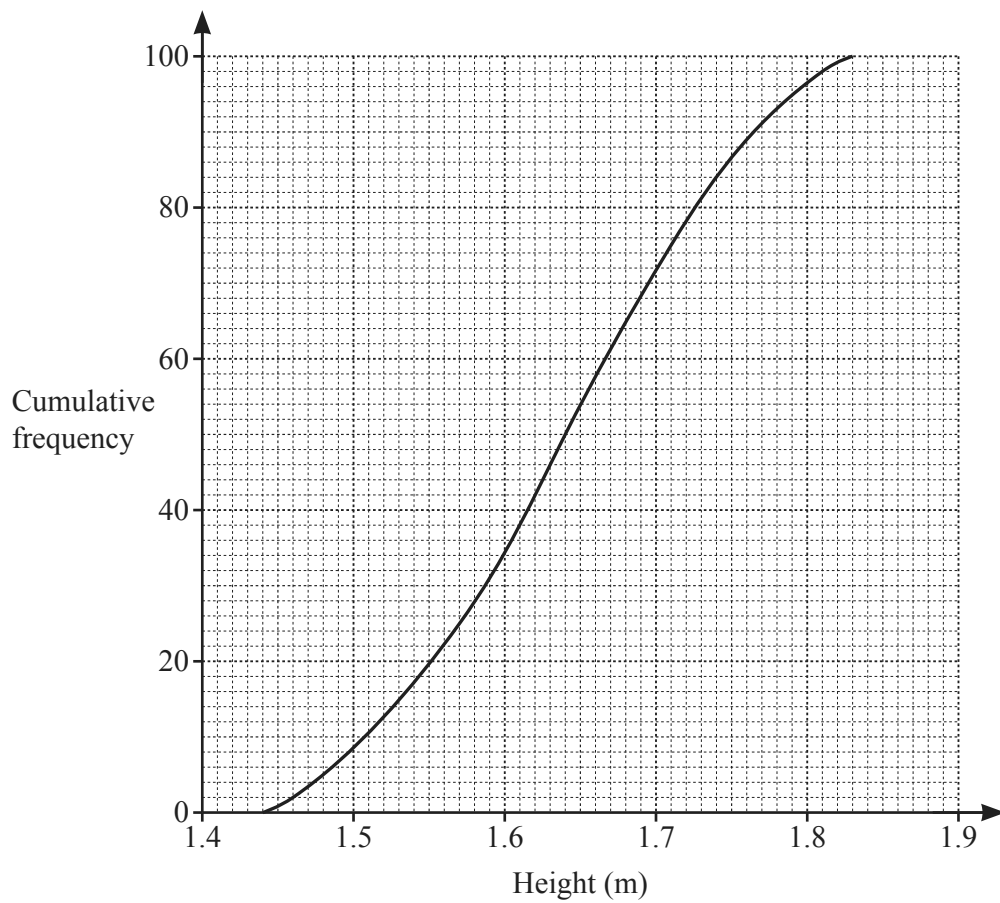


[3]

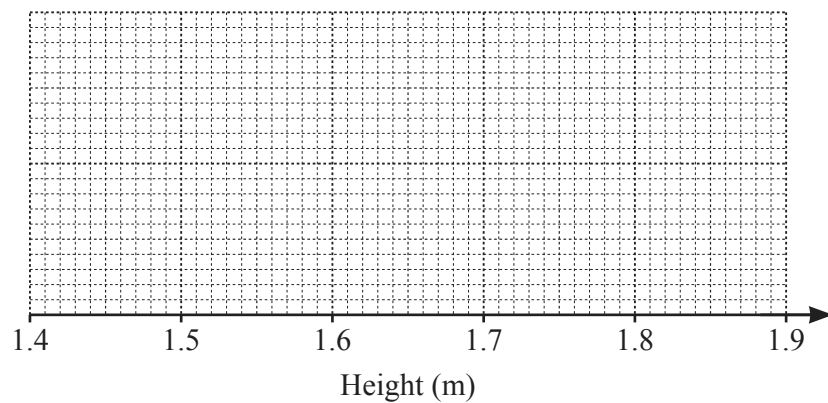
- (iii) Use your diagram to find an estimate of the number of children with a mass of 32 kg or less.

..... [1]

- (b) This cumulative frequency diagram shows information about the height, in metres, of each of 100 students.

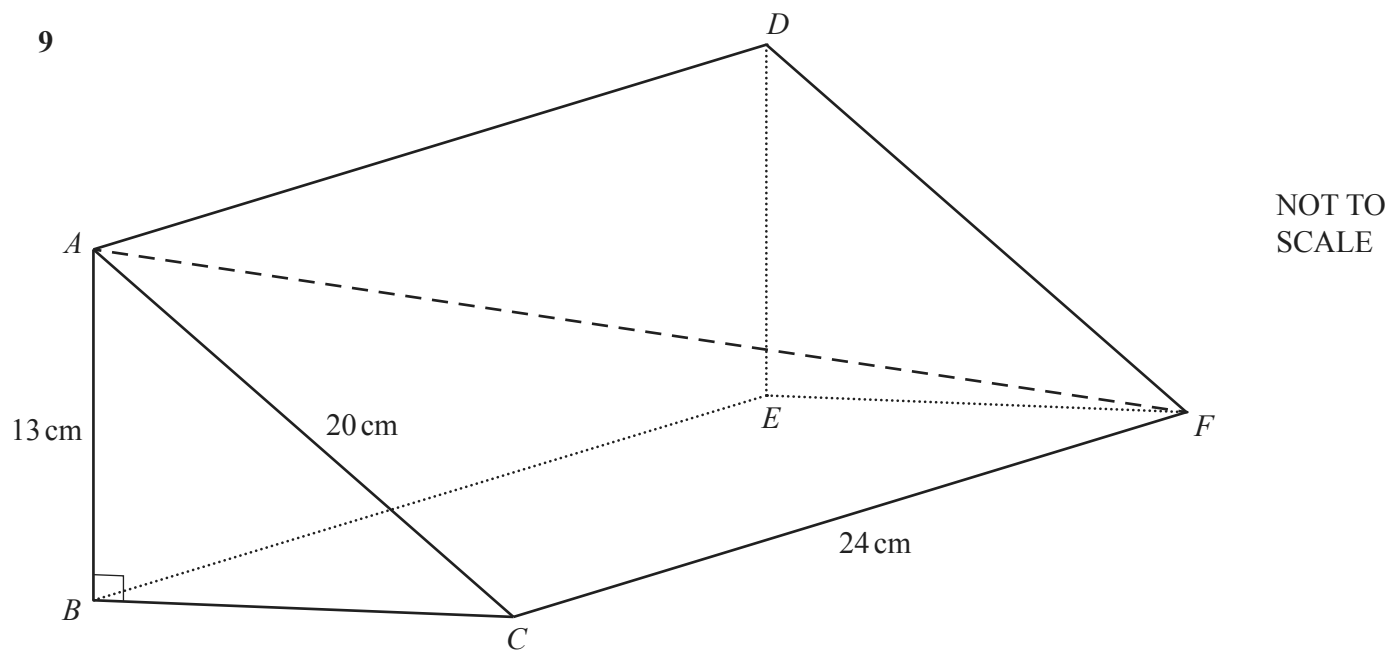


The height of the tallest student is 1.83 metres.
The height of the shortest student is 1.45 metres.



On this grid, draw a box-and-whisker plot for the heights of the 100 students.

[4]



The diagram shows a prism, $ABCDEF$.

$AB = 13$ cm, $AC = 20$ cm, $CF = 24$ cm and angle $ABC = 90^\circ$.

(a) Calculate the total surface area of the prism.

..... cm^2 [6]

(b) Calculate the volume of the prism.

..... cm^3 [1]

(c) Calculate the angle that AF makes with the base $BCFE$.

..... [4]

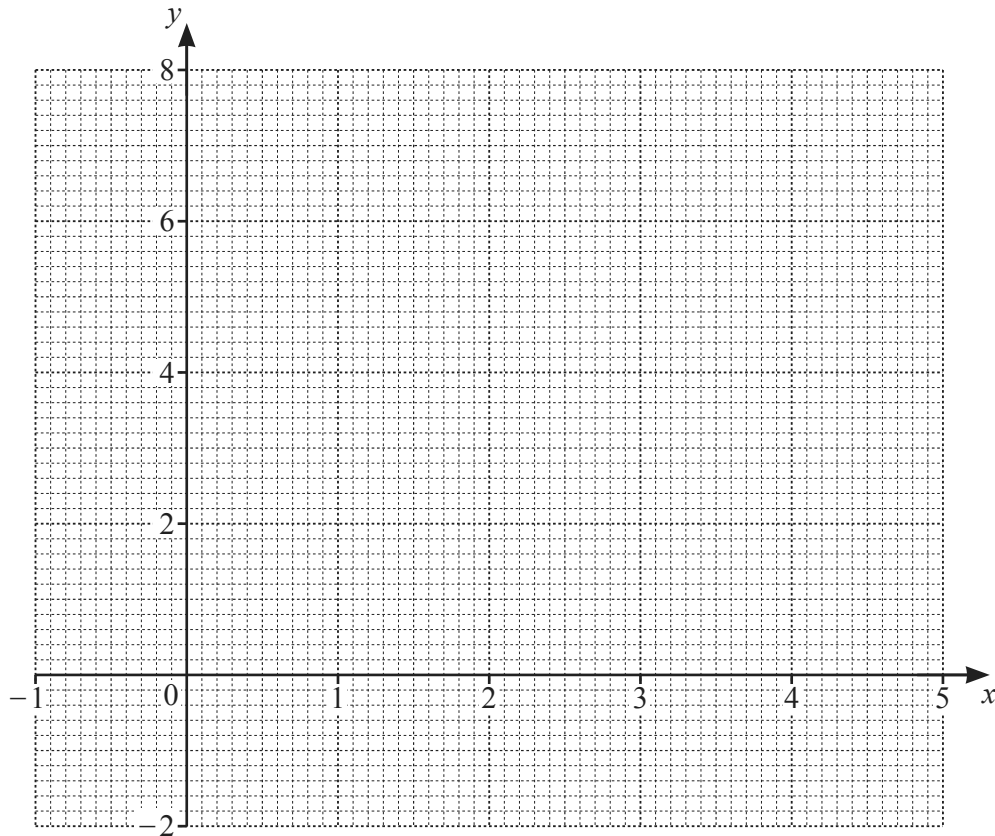
- 10 The table shows some values of $y = 3 + 4x - x^2$ for $-1 \leq x \leq 5$.

x	-1	-0.5	0	1	2	3	4	4.5	5
y	-2			6		6			-2

(a) Complete the table.

[3]

(b) On the grid, draw the graph of $y = 3 + 4x - x^2$ for $-1 \leq x \leq 5$.



[4]

(c) Write down an **integer** value of k for which the equation $3 + 4x - x^2 = k$ has no solutions.

..... [1]

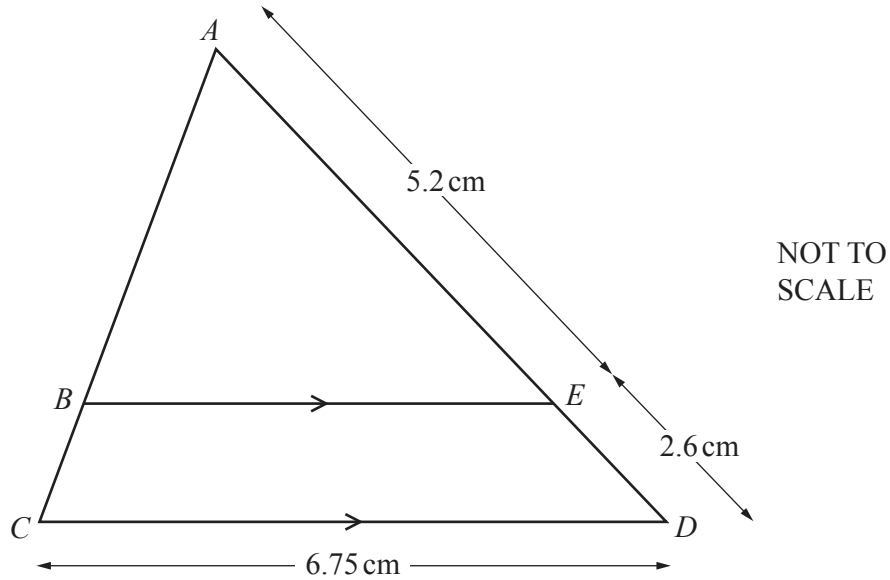
(d) By drawing a suitable straight line on the grid, solve the equation $-1 + \frac{9}{2}x - x^2 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

- 11 (a) Find the size of an exterior angle of a regular polygon with 18 sides.

..... [2]

(b)



In triangle ACD , B lies on AC and E lies on AD such that BE is parallel to CD .
 $AE = 5.2 \text{ cm}$ and $ED = 2.6 \text{ cm}$.

Calculate BE .

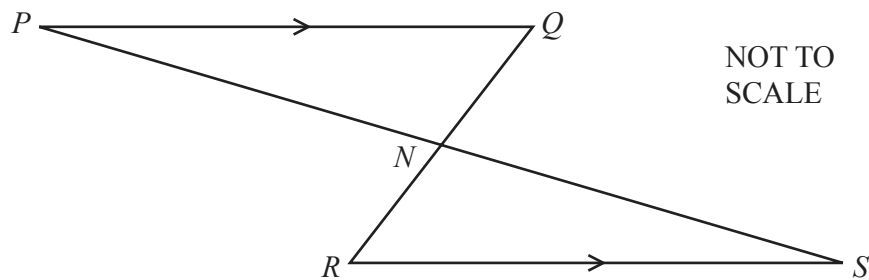
$BE =$ cm [2]

- (c) Two solids are mathematically similar.
 The smaller solid has height 2 cm and volume 32 cm^3 .
 The larger solid has volume 780 cm^3 .

Calculate the height of the larger solid.

..... cm [3]

(d)



PQ is parallel to RS , PNS is a straight line and N is the midpoint of RQ .

Explain, giving reasons, why triangle PQN is congruent to triangle SRN .

.....

 [4]

12 $f(x) = 3 - 2x$ $g(x) = x^2 + 5$ $h(x) = x^3$

(a) Find $f(-5)$.

..... [1]

(b) Find $ff(x)$.
Give your answer in its simplest form.

..... [2]

(c) Solve $g(x) = f(x) + 37$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

(d) Find $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

(e) Find $hf(x) + g(x)$.
Give your answer in its simplest form.

$$\dots\dots\dots [5]$$

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.